

Program : Diploma in Instrumentation Engineering	
Course Code : 6082B	Course Title: Environmental Instrumentation
Semester : 6	Credits: 4
Course Category: Program Elective	
Periods per week : 4 (L:3 T:1 P:0)	Periods per semester: 60

Course Objectives:

- To discuss various sources of pollution.
- To introduce the instrumentation methodologies for environment monitoring.
- To deal with water quality monitoring and waste water treatment.
- To discuss the instrumentation required for air pollution monitoring
- To discuss the instrumentation required for sound pollution
- To introduce instrumentation in weather monitoring

Course Prerequisites:

Topic	Course code	Course Name	Semester
Environmental pollution		Environmental science	2

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Explain instrumentation systems for environment monitoring.	8	Understanding
CO2	Describe the water quality parameters and ground water monitoring	19	Understanding
CO3	Explain Waste Water and Flow Monitoring System	20	Understanding
CO4	Explain Air Pollution and Sound Monitoring Systems	11	Understanding
	Series Test	2	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2				3		
CO2	2				3		
CO3	2				3		
CO4	2				3		

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Explain instrumentation systems for environment monitoring.		
M1.01	Describe the Necessity of instrumentation & control for environment monitoring.	4	Understanding
M1.02	Explain the Instrumentation methodologies for environment monitoring.	4	Understanding
Contents: Introduction- Necessity of instrumentation & control for environment monitoring.-sensor requirement for environment monitoring. Instrumentation methodologies: Ultraviolet analyzers, Gas chromatography in environmental analysis, portable & stationary analytical instruments.			
CO2	Describe the water quality parameters and ground water monitoring		
M2.01	Explain the effects of water quality.	5	Understanding
M2.02	Illustrate Water quality parameters:	5	Understanding
M2.03	Describe Water treatment	5	Understanding
M2.04	Explain Ground water monitoring	4	Understanding
	Series Test-I	1	
Contents: Quality of water Standards of raw & treated water -sources of water & their natural quality, effects of water quality. Water quality parameters: Thermal conductivity, detectors, Opacity monitors, pH analyzers -conductivity analyzers Water treatment: Requirement of water treatment –facilities-methods. Ground water monitoring: Level measurement in ground water monitoring wells, laboratory analysis of ground water samples, instrumentation in ground water monitoring, instrumentation in assessment of soil & ground water pollution			

CO3	Explain Flow Monitoring System For Waste Water		
M3.01	Describe Waste Water measurement techniques	7	Understanding
M3.02	Compare various waste water Flow monitoring methods	7	Understanding
M3.03	Outline Rain water harvesting	6	Understanding
Contents: Waste Water and Flow Monitoring System: Automatic waste water sampling, optimum waste water sampling locations and waste water measurement techniques. Instrumentation set up for waste water treatment plant. Latest methods of waste water treatment plants, Chemical Oxygen Demand (COD), Biochemical Oxygen Demand (BOD). Flow monitoring: Non open channel flow measurement, open channel waste water flow measurement. Rain water harvesting: necessity, methods, role of NGOs & municipal corporation			
CO4	Explain Air Pollution and Sound Monitoring Systems		
M4.01	Explain analytical methods for air pollution	3	Understanding
M4.02	Explain Acoustic noise measurement & monitoring methods	4	Understanding
M4.03	Illustrate the working of Instruments in Weather station	4	Understanding
	Series Test-I	1	
Contents: Air Pollution: Definitions, energy environment relationship, importance of air pollution, Air sampling methods & equipments, analytical methods for air pollution studies. Control of air pollution. Sound pollution: basics of sound pollution, its effect to environment. Acoustic noise measurement & monitoring, Instruments in Weather station: Instruments in Weather station like Barometer, Rain gauge, Ceilometer etc., Virtual Instruments in Environmental Engineering Laboratory, Rover Environmental Monitoring station (REMS).			

Text / References:

T/R	Book Title/Author
R1	Walter J Weber, “Physic- chemical processes for water quality control”, Wiley Inter-science Publications 2012
R2	M N Rao and S K S Rao, “ Air pollution”, TMH publications 26th reprint 2007.
R3	Rao, M. N. and Rao, H. V. N, “ Air Pollution”, Tata McGraw Hill Publishing Company Limited, New Delhi, 1989, ISBN-13: 978-0074518717
R4	Kenneth Wark, Cecil F. Warner, Wayne T. Davis, “Air Pollution: Its Origin and Control” , Pearson; 3 edition (13 November 1997), ISBN-13: 978-0673994165
R5	Peany Howard S, Donal R Rowe and George Tacho Banoylous Teddy, “Environmental Engineering”. McGraw Hill Education; First edition (1 July 2017), ISBN-13: 978-
R6	Patrick F. Cunniff, “ Environmental Noise Pollution” , John Wiley & Sons Inc (4 May 1977) , ISBN-13: 978-0471189435
R7	Gilber M Masters , “Environmental Engineering and Science”, Pearson Education
R8	Randy D. Down & Jay H. Lehr, “Environmental Instrumentation & Analysis Handbook”, Wiley-Blackwell (7 October 2004), ISBN-13: 978-0471463542